Examiner reports

Q1.

Generally this question was answered very well. The most common difficulties were with part (c). One error that appeared here was to use the equation $s = \frac{1}{2}(u + v)t$ with either u or v taken as zero. Often candidates who made errors in part (c) were able to gain the follow through marks in part (d).

Q2.

Most candidates made good attempts at this question. In part (b) (ii) there were some solutions that did not contain enough working to justify the award of marks, for example $a = \frac{4}{0.5} = 8$ with no explanation of where the 4 came from. Some also simply found a way to get 8, for example from something like 9.8 – 2 × 0.9 = 8.

For part (b) (iii) there were many correct solutions, but some candidates took the acceleration as 9.8 instead of 8. The final part proved to be more difficult for the candidates. While there were many candidates who produced good answers, there were also quite a lot who could not give a clear or relevant explanation.

Q3.

Parts (a) and (b) of this question were done very well and many candidates gained all four marks.

Part (c) proved to be more challenging. In part (c)(i), many candidates did not show how they started their working. An equation based on $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ was expected, but not always seen. Some candidates obtained the printed answer, but did not gain full marks as their starting point was not clear. Having both the acceleration and the initial velocity given in the question enabled many candidates to write down a correct expression for the velocity of the particle. Part (c)(iii) was found to be very challenging. If candidates were able to write down an expression for the speed of the particle they would often go on to find the times correctly. It was this vital first step that many candidates were unable to take. A few candidates made errors solving the quadratic equation once they had obtained it correctly.

Q4.

The candidates seemed to be helped by the printed answer in part (a), but some obtained the printed result from the use of equations such as $5 = \frac{1}{2}(-9.8)t^2$. There were also some candidates who did not justify all of their working. There were a number of candidates who found it difficult to work with the two components of the velocity in parts (b), (c) and (d).

A reasonably common error was to use 5 instead of 15 in part (b) and then 15 instead of 5 in part (c). Some candidates also stopped after obtaining the vertical component in part (c) and did not go on to find the speed. Those who did find the components correctly were usually able to go on to find the angle correctly. Many candidates gained the two marks in part (e), sometimes without gaining any others in this question.

Q5.

The candidates generally did well with parts (a) and (b), which were fairly standard, but had more difficulties with the later parts. The main reasons that candidates lost marks in part (a) were because they made errors with the signs in the equations or because they did not form an equation for each particle. Those with sign errors in part (a) often did not obtain the correct tension in part (b).

In part (c), many errors were due to the candidates using the wrong values in the wrong places. For example, in part (c)(i), some used an acceleration of 9.8 instead of 1.96 while in part (c)(ii), some used an acceleration of 1.96 instead of 9.8. Similar confusion concerned the use of the distance of 4 metres in part (c)(i) instead of in part (c)(ii), while the time of 2 seconds was sometimes used in part (c)(ii) instead of in part (c)(i).

Q6.

The majority of the candidates were able to find the initial velocity as requested in part (a), but most found the rest of the question difficult. A reasonable number of the candidates were able to find the acceleration, some by calculating two velocities and using $\mathbf{v} = \mathbf{u} + \mathbf{a}t$, but others clearly could not find a strategy to find the acceleration. Part (c) was found to be more demanding. Some candidates did use the components correctly, but quite a number of candidates did not recognise the approach that was required.

Q7.

This question was done well by the vast majority of candidates, with many scoring full marks. Although not made often, the most common error was failing to multiply by 0.5, or divide by 2, when finding the distances in parts (b) and (c). Most candidates used a correct approach to find the average speed in part (d). Another error, seen in part (e), was to find 2400 but not subtract the answer to part (c).

Q8.

The candidates generally did well with parts (a) and (b), which were fairly standard, but had more difficulties with the later parts. The main reasons that candidates lost marks in part (a) were because they made errors with the signs in the equations or because they did not form an equation for each particle. Those with sign errors in part (a) often did not obtain the correct tension in part (b).

In part (c), many errors were due to the candidates using the wrong values in the wrong places.

For example, in part (c)(i), some used an acceleration of 9.8 instead of 1.96, while in parts (c)(ii) and (c)(iii), some used an acceleration of 1.96 instead of 9.8. Similar confusion concerned the use of the distance of 4 metres in part (c)(i) instead of the later parts, while the time of 2 seconds was sometimes used in the later parts instead of in part (c)(i).

In part (c)(iii), the most common approaches involved the use of constant acceleration equations. While some candidates used the correct numerical values in their equations, many made errors with the signs. Another common approach was to find two times and add them together. With this method also, sign errors were often made by the candidates.

Q9.

While some candidates did well with this question, others made some fundamental errors. In part (a), some candidates found the magnitude of the acceleration and then tried to use this scalar quantity in the later parts of the question, ending up with a mix of scalar and vector quantities.

In part (b)(i), many candidates found the position vector, but did not go on to find the distance as requested. There were quite a number of correct responses to part (b)(ii), but quite often this was not followed by a correct approach to the final part of the question.

Q10.

Very many candidates were able to prove the result in part (a). Most considered the vertical component of the velocity and found the result easily. Some tried to find the time of flight first. In some cases these candidates halved their answer, but some did not do this or did not make it clear. A very small number of candidates used equations that they had learned for the time of flight or similar, but did not justify these results and so did not gain the marks. Candidates should not use learned formulae for the range etc in questions like this.

There were many different approaches to part (b)(i). Quite often candidates could produce a correct equation as a starting point, but failed to simplify it correctly. Often the candidates ended up with a mixture of $\sin \alpha$ and $\sin^2 \alpha$ terms. There were many minor arithmetic or algebraic errors. In part (b)(ii), many candidates used the incorrect time: their approach was based on a time of $\frac{3 \sin \alpha}{2}$ instead of $3 \sin \alpha$.

Part (c) was often done well, but quite a few candidates incorrectly talked about the projectile having no mass.

Q11.

There were many good solutions to part (a). Only a very small number of candidates ignored the instruction to form an equation of motion for each particle. A few candidates assumed that the heavier particle would accelerate upwards when released. Some of these candidates worked through to produce a negative acceleration and stated that magnitude would be the required value. However, some of the candidates who took this approach “lost” a negative sign in their working and ended up with a positive acceleration.

In part (b) there was one common error, which was seen on a fairly large number of scripts. This was to obtain the distance moved by each particle, but not to double this to give the distance between the particles. A very small number of candidates used an acceleration of 9.8 instead of 0.98.

Q12.

This question also provided more challenge for the candidates. In this question a number of candidates did assume that the acceleration due to gravity was positive.

In part (a), quite a number of candidates found the time of flight and then used half of this value to calculate the maximum height. There were however a number of cases where the candidates did not half their value for the time of flight and so produced solutions which did not relate to the maximum height.

Part (b) was much more demanding and relatively few correct solutions were seen. Those candidates who identified the required method were able to do well, but the majority made no attempt or produced working which could not lead to the required answer. Similarly part (c) was found to quite difficult, but some candidates did use the answer from part (b) to put together correct solutions for part (c) when they had been unable to do part (b).

Q13.

Overall this question was done well, but there were a few issues that appeared with a fair number of scripts. In part (a) some candidates often gave only two times and some gave the times when the speed was a minimum or maximum. Part (b) provided examples of candidates showing insufficient working. For example, some candidates simply wrote $5 \times 15 = 75$, which did not provide enough justification to gain any marks. Parts (c) and (d) were generally done very well with the occasional arithmetic error.

Q14.

Parts (a) and (b), about the assumptions, were generally answered well. The most common incorrect answer was to say that the peg was fixed in part (a).

There were many good solutions to part (c). Only a very small number of candidates ignored the instruction to form an equation of motion for each particle. A few candidates assumed that the heavier particle would accelerate upwards when released. Some of these candidates worked through to produce a negative acceleration and stated that magnitude would be the required value. However, some of the candidates who took this approach "lost" a negative sign in their working and ended up with a positive acceleration.

In part (d) there was one common error, which was seen on a fairly large number of scripts. This was to obtain the distance moved by each particle, but not to double this to give the distance between the particles. A very small number of candidates used an acceleration of 9.8 instead of 0.98.

Q15.

Parts (a) and (b) were generally done well by the candidates. A few candidates subtracted the vectors in part (a) instead of adding them. Also some tried to find the magnitude of each vector in part (a) and then lost several marks because they had very confused working.

In part (c) there were many attempts which involved the division of vectors. For example working such as: $m = \dfrac{9\mathbf{i} + 12\mathbf{j}}{1.5\mathbf{i} + 2\mathbf{j}} = 6\mathbf{i} + 6\mathbf{j}$. Candidates should be advised to consider the components involved and not to attempt to divide vectors.

There were a number of errors that appeared in part (d). One was to include an initial velocity, which was equal to the resultant force on the particle. A few candidates used incorrect formulae. In the final part of the question a large number of candidates gave the vector $3\mathbf{i} + 4\mathbf{j}$ as their final answer and did not find the magnitude in order to get the distance as requested.

Q16.

Parts (a) and (c) were done well, with almost all of the candidates gaining full marks on these parts. Part (b) was a little more demanding. There were a number of correct solutions, but the two most common errors were to subtract 400 from 660 to obtain 260 N or to include the weight of the motorcycle and rider in some way.

Q17.

Part (a) of this question was done very well, but part (b) caused difficulties for some candidates. The most common errors in part (b) were to include the weight of the car and trailer or to produce equations in which the wrong combinations of forces were used.

Q18.

There were many very good responses to this question. Generally, the candidates who did have difficulties with this question had problems in part (a) but were able to complete part (b) and gain the follow-through marks by using their answer to part (a). The sort of difficulties encountered in part (a) included:

- using a value for the acceleration, for example g or zero in their constant acceleration equation;
- simply using $u = \frac{s}{t}$;
- using $s = \frac{1}{2}(u+v)t$, but then having difficulty solving for u.

Q19.

There were lots of good responses to this question, although part (e) was sometimes incorrect.

Part (a) was found to be very straightforward and the majority of the candidates gained the marks, even if they did not go on to do other parts of the question.

Part (b) was also done well, with the vast majority of candidates stating clearly the two equations of motion that they went on to solve. A few did use a method with only one equation, but fortunately this method was not common. Also, a few used the weight of the block instead of the friction force.

Part (c) was also done well, provided the candidates had an appropriate equation to use from part (b).

It was interesting that some candidates who had found parts (a) to (c) difficult were able to use the acceleration given and produce correct answers to parts (d) and (e). There were many good solutions to part (d), but in part (e) candidates often continued to use the acceleration as 1.225 m s^{-2} or took the initial velocity of the particle to be zero. Some candidates made both of these errors. The success rate for part (e) was lower than that for part (d).

Q20.

This question was demanding for the candidates, although part (a) caused very few difficulties for the majority, and some gained only the three marks for this part of the question.

Part (b) was the most demanding part of the question, and only very few candidates gained the marks available. Many candidates stated that the **i** component should be zero, but very few formed an equation to find the time when this would be. Most incorrect approaches were based on using a time, often 20 seconds, and then calculating the **j** component of the velocity and stating that the **i** component was 0.

There were some good answers to part (c), but quite a few candidates failed to include the initial position of the particle.

In part (d), there were some good solutions, but some candidates made errors. These included not subtracting the initial position. Some candidates used $\frac{\mathbf{u}+\mathbf{v}}{2}$. With this method there were two main errors: dividing by 20 rather than 2 and using an incorrect expression for either **u** or **v**.

Q21.

Most candidates did well on this question, having few difficulties finding the magnitude of the friction force. In part (b), the great majority of candidates took a correct approach to finding the acceleration. There were a few candidates who did not consider the block and the particle as two separate bodies. Part (c) was done well with candidates experiencing very few difficulties.

Q22.

While there were many good responses to this question, candidates did make some errors and some did not show enough working in part (a). In part (b), there were a number of incorrect approaches. These included:

- Simply calculating the weight.
- Calculating the magnitude of the resultant force using $F = ma$.
- Making a sign error by producing an equation such as $70g - T = 70 \times 0.64$.

Part (c) was usually done well, although a few candidates did not appear to know how to calculate the average speed.

Q23.

Generally part (a) was done well, whilst part (b) caused more difficulties. The fact that the acceleration was given did seem to help many candidates, but some did not justify the minus sign. Applying the constant acceleration equation to find the distance was also done well, although a few candidates took $u = 0$ and $v = 4$. There were a number of candidates who used two constant acceleration equations, finding the time taken to stop as an intermediate step. A few of these candidates did not go on to find the distance.

In part (b), many of the candidates were able to find the magnitude of the friction force. In a few cases the candidates did not show how the reaction force was obtained. Full marks were not awarded if the candidates simply stated $R = 1.84$ without any justification. When calculating the acceleration there were many sign errors in candidates' equations and some candidates also included a”4” which appeared as if it was a force. While many candidates realised that the puck would remain at rest, only a relatively small number of them were able to give a convincing explanation. Many of the explanations contained vague statements and lacked precision and clarity.

Q24.

In this question the use of vector notation by some candidates was quite poor. Part (a) was done well by a good number of candidates, but some did not show enough working to gain full marks. Typical issues were not showing that their equation was derived from $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ and not showing the division by 40 clearly. Some candidates did well with part (b), but there were quite a lot of arithmetic errors.

Some candidates used an incorrect initial velocity, for example 4**i** instead of 5**j**. Part (c) proved to be too demanding for the majority of the candidates, although some were able to gain a few marks. One of the most common errors was to work with position vectors rather than with velocities. For those who did work with velocities, creating an equation to find the time when the Jet Ski was travelling south east was found to be difficult. Quite a number of the candidates who got this far had equations that lacked a minus sign.

Q25.

This question was done very well by the vast majority of the candidates. Almost all gained full marks on part (a), but a few did make arithmetic errors. Again, in part (b), almost all the candidates found the acceleration correctly, although a few did give the answer $\frac{4}{3}$. Part (c) did cause a little more difficulty, although there were many correct responses. The two most common errors were either to assume that the tension was equal to the weight or to make a sign error and obtain an equation such as
$400g - T = 400a$.

Q26.

Generally the candidates made quite good attempts at this question, but some parts caused the candidates more difficulty. A few candidates made poor use of vector notation. Part (a) was answered correctly by the vast majority of the candidates. The most common error was dealing incorrectly with the initial velocity. A few went on to simplify their answers incorrectly. Part (b) was a little more demanding, as some candidates confused the components. Part (c) did cause a little more difficulty for some candidates, with a few candidates not giving any answer at all. However, there were again many correct solutions.

In part (d)(i), although there were many correct solutions, some candidates showed that the helicopter was south or north of the origin, rather than specifically north. Interestingly, some candidates who could not answer part (c) were able to write down the position vector at the specified time. A few candidates found the correct position vector but did not conclude that the helicopter was due north.

When answering part (d)(ii), many of the candidates found the correct velocity, but some did not find the speed as requested.

Q27.

This question was done very well by the vast majority of candidates. Many produced completely correct solutions. There were a number of minor arithmetic or algebraic errors in some solutions.

Q28.

The first two parts of this question were done very well by the vast majority of candidates, but part (c) caused more difficulties. In part (a), the candidates almost always drew a graph with the correct shape, but made errors on the axes. The two most common errors were to use 21 instead of 12 on the time axis and to use an incorrect value on the velocity axis. In part (b), a very small number of candidates made arithmetic errors and a few also used incorrect values.

Part (c) caused more difficulties for the candidates. There were three main incorrect approaches. The first of these was to assume that the forces on the lift were in equilibrium and hence simply state that the tension was equal to the weight, giving their answer as 2940 N. Some candidates used an incorrect value for the acceleration, which they then included in a three term equation of motion which would have otherwise been correct. The most common approach was to take the acceleration as 2 m s^{-2}.

A few candidates used a three term equation of motion with the correct terms but an incorrect sign. This approach would produce a result such as T = 2940 – 150 = 2790 N.

Q29.

Almost all candidates gained some marks on this question, but only a fairly small proportion gained full marks. In part (a), there were many good answers, but a number of candidates stated assumptions that were not about the string. Instead they produced comments about the block or the peg. Part (b) was generally done quite well, but there were some candidates who did not gain credit because they did not show enough working. For example, simply stating 6/4 = 1.5 was not sufficient. There were, in contrast, some very good solutions that fully justified the final answer.

In part (c), the candidates who considered the forces on the 7 kg sphere did quite well. The main error that these candidates made was to use $T - 7g = 7a$, rather than $7g - T = 7a$ and obtained a tension of 79.1 N. Other candidates produced equations that involved the 13 kg block but could make little satisfactory progress because they did not know the friction force.

Part (d) was the most demanding part of the question. It required the candidates to find the friction force and then to use $F = \mu R$. Some candidates worked through these stages, but there were relatively few complete solutions. The most common errors were to assume that the friction was equal to the tension and simply apply $F = \mu R$, and to try to work with a combination of both bodies instead of just the block.

Q30.

Part (a) was usually done very well, but in some cases very badly. Many candidates produced good complete solutions. A few made arithmetic or algebraic errors that often resulted in the loss of the final accuracy mark. A few candidates found the time, but not the height. Part (b) was more demanding. Some candidates did not really make any progress with this part of the question. For those that got started, the most common errors were to include a sign error in the quadratic equation that they used to find the time; and to produce a correct quadratic equation, but to make an error in finding its solutions.

Some candidates did use a two-time approach, finding the time to go up to the maximum height and the time to come down to the tree. Many of these candidates were successful, but there were problems associated with getting the right distance to use to find the time down to the tree.

Q31.

Many candidates made good attempts at parts (a) and (b). The majority of errors were due to errors made manipulating the vectors. Some candidates did write down correct equations or expressions to begin their solutions, but made errors in manipulating their expressions. In part (b), a few candidates included a 75**i** term.

Several candidates simply did not attempt part (c) of the question. Of those that did, some tried to find a time using expressions for the position of the particle rather than its velocity, while other candidates tried to use the equation $v^2 = u^2 + 2as$.

Q32.

This question was generally done very well, particularly parts (a) and (b). In part (c), some candidates simply assumed that the tension would be equal to the weight. Some of these candidates gave the same answer for part (d) and gained the mark there. A few candidates performed the calculation 4410 − 45 and obtained an incorrect value.

Q33.

Parts (a) and (b) of this question were done well by many candidates, but part (c) was more demanding. In part (a) most candidates formed two equations as requested, but a few only worked with one equation for the whole system. Those who had done part (a) correctly had few problems with part (b), while those with incorrect equations in part (a) obtained incorrect answers as a result. The main problem in part (c) was that many candidates assumed that the particles had moved 1 m each, instead of the 0.5 m that was needed for them to be 1 m apart. A few candidates took the acceleration to be 9.8 m s^{-2} in this part of the question.

Q34.

This question caused problems for a number of candidates. The most common error, which created difficulties throughout the question, was to treat the ball as if it had an initial vertical component of velocity, often equal to 20 m s^{-1}. This would lead to equations of the form $2.45 = 4.9t^2 \pm 20t$. A few candidates who were unable to do part (a) were able to use the printed answer from part (a) to complete part (b) correctly, but some did include an initial vertical component of velocity again.

Part (c) of this question was more demanding. There were two main errors. The first was to work with the position of the ball when it hit the ground. These candidates would calculate an angle based on the two distances that they knew. A typical answer for these candidates was $\theta = \tan^{-1}\left(\frac{2.45}{14.14}\right)$. The other error was to only find one component of the ball's velocity.

Q35.

Generally this question was very well done. However some candidates did lose marks in part (a) for simply stating “$9.8 \times 1.5 = 14.7$” without any supporting work to justify this calculation. In part (c), some candidates took a fairly lengthy approach which involved finding the time first, but usually leading to a correct solution. Some candidates used $t = 1.5$, and hence obtained incorrect solutions.